

First Derivatives and Convex Function

Learning objectives

By the end of this chapter, you should be able to

- explain why every local minimizer of a convex problem is global;
- compute gradients, directional derivatives, and derivatives along curves;
- relate gradients to level sets and identify Euclidean steepest directions;
- prove and apply the tangent-plane characterization of differentiable convex functions;
- use the first-order variational inequality to characterize minimizers over a convex set.

Convexity turns local information into global information. A local minimum cannot be a misleading “valley” when both the objective and feasible geometry are convex. When derivatives are available, the defining chord inequality can be replaced by an inequality involving a tangent plane, leading to a practical optimality condition.

4.1 Local minimizers become global

Theorem 4.1 (Local-to-global principle). *Let $C \subseteq \mathbb{R}^n$ be nonempty and convex, and let $f : C \rightarrow \mathbb{R}$ be convex. Every local minimizer of f over C is a global minimizer. If f is strictly convex, a global minimizer, if one exists, is unique.*

Proof. Let $\mathbf{x}^*$ be a local minimizer. There is a radius $\rho > 0$ such that

$$f(\mathbf{x}^*) \leq f(\mathbf{x}) \quad \text{for every } \mathbf{x} \in C \text{ satisfying } \|\mathbf{x} - \mathbf{x}^*\| < \rho.$$

Suppose, for a contradiction, that $\mathbf{x}^*$ is not global. Then there is $\tilde{\mathbf{x}}^* \in C$ such that $f(\tilde{\mathbf{x}}^*) < f(\mathbf{x}^*)$. Define

$$\delta := \min \left\{ \frac{\rho}{2}, \frac{\|\tilde{\mathbf{x}}^* - \mathbf{x}^*\|}{2} \right\}, \quad \theta := \frac{\delta}{2\|\tilde{\mathbf{x}}^* - \mathbf{x}^*\|}, \quad \mathbf{y} := (1 - \theta)\mathbf{x}^* + \theta\tilde{\mathbf{x}}^*.$$

The strict objective inequality ensures $\tilde{\mathbf{x}}^* \neq \mathbf{x}^*$, so these quantities are well defined and $\theta \in (0, 1)$. Convexity of C gives $\mathbf{y} \in C$, while $\|\mathbf{y} - \mathbf{x}^*\| = \delta/2 < \rho$. Local optimality therefore gives $f(\mathbf{y}) \geq f(\mathbf{x}^*)$. On the other hand, convexity of f gives

$$f(\mathbf{y}) \leq (1 - \theta)f(\mathbf{x}^*) + \theta f(\tilde{\mathbf{x}}^*) < f(\mathbf{x}^*),$$

a contradiction. Hence $\mathbf{x}^*$ is global.

For uniqueness, suppose that $\mathbf{x}^*$ and $\tilde{\mathbf{x}}^* \neq \mathbf{x}^*$ are both global minimizers. Define $\mathbf{w} := \frac{1}{2}(\mathbf{x}^* + \tilde{\mathbf{x}}^*)$. Convexity of C gives $\mathbf{w} \in C$, whereas strict convexity gives

$$f(\mathbf{w}) < \frac{1}{2}f(\mathbf{x}^*) + \frac{1}{2}f(\tilde{\mathbf{x}}^*) = f(\mathbf{x}^*),$$

contradicting global optimality. $\square$

Warning

Strict convexity is sufficient, but not necessary, for uniqueness. For example, $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) := |x|$ is not strictly convex but has the unique minimizer 0. Strict convexity also does not guarantee existence: $g : \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(x) := e^x$ is strictly convex, but $\inf_{x \in \mathbb{R}} g(x) = 0$ is not attained.

4.2 Gradients and directional derivatives

Definition 4.1 (Interior point). Let $S \subseteq \mathbb{R}^n$. A point $\mathbf{x}_0 \in S$ is an *interior point* of S if there exists $\varepsilon > 0$ such that

$$\|\mathbf{y} - \mathbf{x}_0\| < \varepsilon \implies \mathbf{y} \in S \quad \text{for every } \mathbf{y} \in \mathbb{R}^n.$$

Definition 4.2 (Gradient). Let $S \subseteq \mathbb{R}^n$, let $f : S \rightarrow \mathbb{R}$, and let $\mathbf{x} \in S$ be an interior point. If f is differentiable at $\mathbf{x}$, its gradient is the column vector

$$\nabla f(\mathbf{x}) := \begin{bmatrix} \frac{\partial f}{\partial x_1}(\mathbf{x}) \\ \vdots \\ \frac{\partial f}{\partial x_n}(\mathbf{x}) \end{bmatrix}.$$

Proposition 4.2 (Directional derivatives and curves). Let $S \subseteq \mathbb{R}^n$, let $f : S \rightarrow \mathbb{R}$, and suppose f is differentiable at an interior point $\mathbf{x} \in S$.

(a) For every $\mathbf{d} \in \mathbb{R}^n$, the directional derivative is

$$Df(\mathbf{x}; \mathbf{d}) := \lim_{t \rightarrow 0} \frac{f(\mathbf{x} + t\mathbf{d}) - f(\mathbf{x})}{t} = \nabla f(\mathbf{x})^\top \mathbf{d}.$$

(b) Let $\delta > 0$, let $\gamma : (-\delta, \delta) \rightarrow S$ be differentiable at 0, and suppose $\gamma(0) = \mathbf{x}$. Define $g : (-\delta, \delta) \rightarrow \mathbb{R}$ by $g := f \circ \gamma$. Then

$$g'(0) = \nabla f(\mathbf{x})^\top \gamma'(0).$$

Proof. For $\mathbf{h}$ sufficiently close to $\mathbf{0}$, define the remainder

$$r(\mathbf{h}) := f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) - \nabla f(\mathbf{x})^\top \mathbf{h}.$$

Differentiability means precisely that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{r(\mathbf{h})}{\|\mathbf{h}\|} = 0.$$

For a fixed nonzero $\mathbf{d}$, set $\mathbf{h} := t\mathbf{d}$. Then

$$\frac{f(\mathbf{x} + t\mathbf{d}) - f(\mathbf{x})}{t} = \nabla f(\mathbf{x})^\top \mathbf{d} + \frac{r(t\mathbf{d})}{t},$$

and $|r(t\mathbf{d})/t| = |r(t\mathbf{d})| \|\mathbf{d}\| / \|t\mathbf{d}\| \rightarrow 0$. The case $\mathbf{d} = \mathbf{0}$ is immediate, proving part (a).

For part (b), differentiability of γ at 0 means that the curve remainder

$$\boldsymbol{\eta}(t) := \gamma(t) - \mathbf{x} - t\gamma'(0)$$

satisfies $\|\boldsymbol{\eta}(t)\| / |t| \rightarrow 0$. Substitute $\mathbf{h} := \gamma(t) - \mathbf{x} = t\gamma'(0) + \boldsymbol{\eta}(t)$ into the preceding expansion, divide by t , and let $t \rightarrow 0$. The two remainder terms vanish, leaving $g'(0) = \nabla f(\mathbf{x})^\top \gamma'(0)$. $\square$

If $\mathbf{d}$ is a unit vector, $\nabla f(\mathbf{x})^\top \mathbf{d}$ is the rate of change per unit Euclidean distance. Cauchy–Schwarz gives

$$-\|\nabla f(\mathbf{x})\| \leq \nabla f(\mathbf{x})^\top \mathbf{d} \leq \|\nabla f(\mathbf{x})\|.$$

When $\nabla f(\mathbf{x}) \neq \mathbf{0}$, the right-hand and left-hand equalities occur at

$$\mathbf{d}_{\text{asc}} := \frac{\nabla f(\mathbf{x})}{\|\nabla f(\mathbf{x})\|}, \quad \mathbf{d}_{\text{desc}} := -\frac{\nabla f(\mathbf{x})}{\|\nabla f(\mathbf{x})\|},$$

respectively. Thus $\mathbf{d}_{\text{asc}}$ and $\mathbf{d}_{\text{desc}}$ are the Euclidean unit directions of steepest ascent and descent. If $\nabla f(\mathbf{x}) = \mathbf{0}$, every directional derivative is zero, so there is no unique first-order steepest direction.

Warning

The word “steepest” is meaningful only after the size of a direction has been fixed. The formulas above compare directions of unit Euclidean norm. With another norm, the steepest direction can change.

Example 4.1 (Linear and quadratic gradients). Let $\mathbf{b} \in \mathbb{R}^n$, and define $f : \mathbb{R}^n \rightarrow \mathbb{R}$ by $f(\mathbf{x}) := \mathbf{b}^\top \mathbf{x}$. Then $\nabla f(\mathbf{x}) = \mathbf{b}$. If $A \in \mathbb{R}^{n \times n}$ is symmetric and $q : \mathbb{R}^n \rightarrow \mathbb{R}$ is defined by

$$q(\mathbf{x}) := \frac{1}{2} \mathbf{x}^\top A \mathbf{x},$$

then $\nabla q(\mathbf{x}) = A\mathbf{x}$.

4.3 Level-set geometry

Let $D \subseteq \mathbb{R}^n$, let $f : D \rightarrow \mathbb{R}$, and let $\alpha \in \mathbb{R}$. The set

$$E_\alpha(f) := \{\mathbf{x} \in D : f(\mathbf{x}) = \alpha\}$$

is the *level set* of f at level α . When $n = 2$, a level set is also called a *level curve* or *contour*.

Proposition 4.3 (Gradient normality). *Let $\mathbf{x} \in E_\alpha(f)$ be an interior point of D , and suppose that f is differentiable at $\mathbf{x}$ with $\nabla f(\mathbf{x}) \neq \mathbf{0}$. For $\delta > 0$, let $\gamma : (-\delta, \delta) \rightarrow E_\alpha(f)$ be differentiable at 0 and satisfy $\gamma(0) = \mathbf{x}$. Then*

$$\nabla f(\mathbf{x})^\top \gamma'(0) = 0.$$

The vector $\gamma'(0)$ is the tangent direction traced by the curve, so the nonzero vector $\nabla f(\mathbf{x})$ is normal to that direction. Figure 4.1 gives an illustration.

Proof. Define $g : (-\delta, \delta) \rightarrow \mathbb{R}$ by $g := f \circ \gamma$. Since $g(t) = \alpha$, Proposition 4.2 gives

$$0 = g'(0) = \nabla f(\mathbf{x})^\top \gamma'(0).$$

□

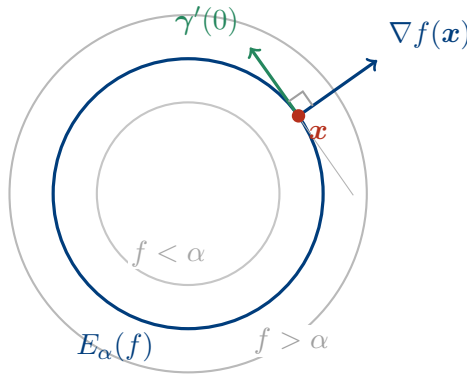


Figure 4.1: The gradient is normal to a differentiable level-set curve at a point where the gradient is nonzero.

Warning

Do not confuse the level set $E_\alpha(f)$ with the *sublevel set*

$$L_\alpha(f) := \{\mathbf{x} \in D : f(\mathbf{x}) \leq \alpha\}.$$

Moreover, orthogonality to the gradient is necessary for a curve tangent, but under mere differentiability it need not characterize all tangents.

4.3.1 A preview of zig-zagging

Because the negative gradient is normal to the current level curve, repeated steps across elongated contours can zig-zag. Figure 4.2 previews the phenomenon; the method and its exact-search orthogonality are developed in Chapter 10.

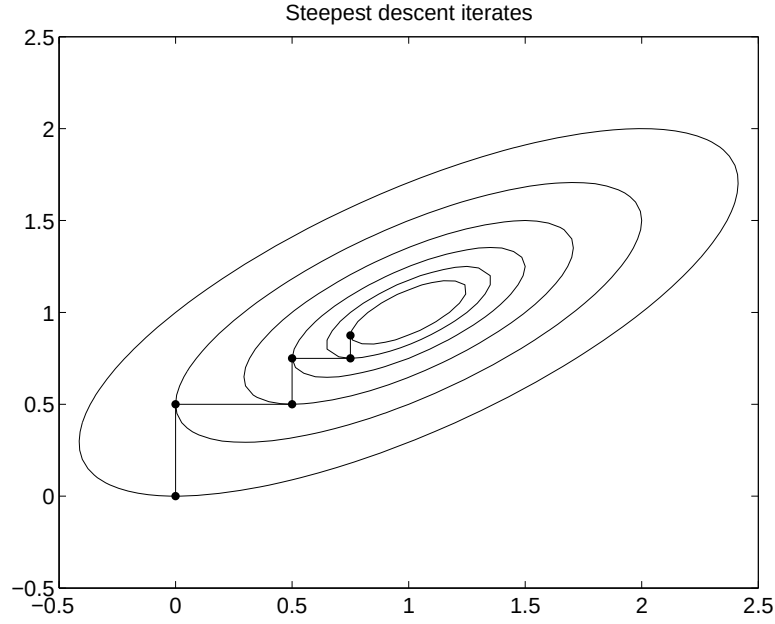


Figure 4.2: A preview of the zig-zag path generated by successive steepest-descent directions.

Example 4.2 (A parabolic level curve). Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $f(x_1, x_2) := x_1 - x_2^2$. Then

$$\nabla f(x_1, x_2) = (1, -2x_2)^\top.$$

Define $\gamma : \mathbb{R} \rightarrow E_1(f)$ by

$$\gamma(t) := (1 + (1+t)^2, 1+t)^\top.$$

Then $\gamma(0) = (2, 1)^\top$ and $\gamma'(0) = (2, 1)^\top$. At this point,

$$\nabla f(2, 1)^\top \gamma'(0) = (1, -2)(2, 1)^\top = 0.$$

Thus $(1, -2)^\top$ is normal to the level curve $x_1 - x_2^2 = 1$ and points toward increasing level values. See the illustration in [Figure 4.3](#).

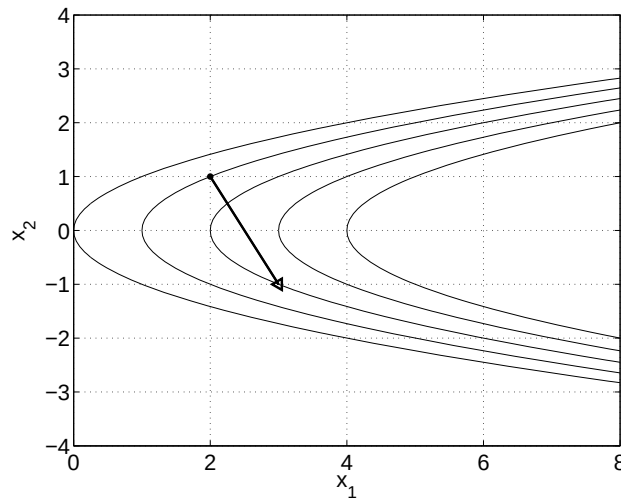


Figure 4.3: Contours of $f(x_1, x_2) = x_1 - x_2^2$, with a normal direction at $(2, 1)$.

4.4 Tangent-plane characterization of convexity

Let $C \subseteq \mathbb{R}^n$ be open, let $f : C \rightarrow \mathbb{R}$ be differentiable, and fix $\mathbf{x} \in C$. Define the affine function $\ell_{\mathbf{x}} : C \rightarrow \mathbb{R}$ by

$$\ell_{\mathbf{x}}(\mathbf{y}) := f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}).$$

It is the first-order affine approximation to f at $\mathbf{x}$. For a convex function, every such tangent plane is a global underestimator; see Figure 4.4.

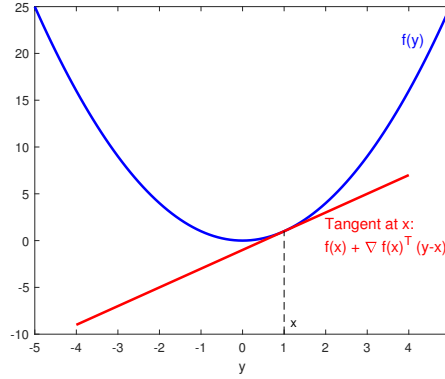


Figure 4.4: The tangent plane to a differentiable convex function lies below its graph.

Theorem 4.4 (Tangent-plane characterization). *Let $C \subseteq \mathbb{R}^n$ be open and convex, and suppose $f : C \rightarrow \mathbb{R}$ is continuously differentiable. Then f is convex if and only if*

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) \quad \forall \mathbf{x}, \mathbf{y} \in C. \quad (4.1)$$

Moreover, f is strictly convex if and only if the inequality is strict whenever $\mathbf{x} \neq \mathbf{y}$.

Proof. Suppose first that f is convex. For $t \in (0, 1]$,

$$f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) = f((1-t)\mathbf{x} + t\mathbf{y}) \leq (1-t)f(\mathbf{x}) + tf(\mathbf{y}).$$

After subtracting $f(\mathbf{x})$ and dividing by t ,

$$\frac{f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{t} \leq f(\mathbf{y}) - f(\mathbf{x}).$$

Letting $t \downarrow 0$ and using Proposition 4.2 proves Equation (4.1).

Conversely, assume Equation (4.1). Let $\mathbf{u}, \mathbf{v} \in C$, $\lambda \in (0, 1)$, and

$$\mathbf{w} := \lambda \mathbf{u} + (1 - \lambda)\mathbf{v}.$$

Apply the tangent inequality at base point $\mathbf{w}$, first with endpoint $\mathbf{u}$ and then with endpoint $\mathbf{v}$:

$$\begin{aligned} f(\mathbf{u}) &\geq f(\mathbf{w}) + \nabla f(\mathbf{w})^\top (\mathbf{u} - \mathbf{w}), \\ f(\mathbf{v}) &\geq f(\mathbf{w}) + \nabla f(\mathbf{w})^\top (\mathbf{v} - \mathbf{w}). \end{aligned}$$

Multiply these inequalities by λ and $1 - \lambda$, respectively, and add. The gradient terms cancel because

$$\lambda(\mathbf{u} - \mathbf{w}) + (1 - \lambda)(\mathbf{v} - \mathbf{w}) = \mathbf{0}.$$

Consequently,

$$f(\lambda \mathbf{u} + (1 - \lambda)\mathbf{v}) = f(\mathbf{w}) \leq \lambda f(\mathbf{u}) + (1 - \lambda)f(\mathbf{v}),$$

so f is convex.

If the tangent inequality is strict for distinct points, the last two endpoint inequalities are strict whenever $\mathbf{u} \neq \mathbf{v}$, proving strict convexity. Conversely, suppose f is strictly convex but equality holds in Equation (4.1) for distinct $\mathbf{x}, \mathbf{y}$. At their midpoint $\mathbf{z} := (\mathbf{x} + \mathbf{y})/2$, strict convexity gives

$$f(\mathbf{z}) < \frac{1}{2}f(\mathbf{x}) + \frac{1}{2}f(\mathbf{y}).$$

The already proved non-strict tangent inequality at $\mathbf{x}$, together with the assumed equality at $\mathbf{y}$, gives

$$f(\mathbf{z}) \geq f(\mathbf{x}) + \frac{1}{2}\nabla f(\mathbf{x})^\top(\mathbf{y} - \mathbf{x}) = \frac{1}{2}f(\mathbf{x}) + \frac{1}{2}f(\mathbf{y}),$$

a contradiction. □

Swapping $\mathbf{x}$ and $\mathbf{y}$ in Equation (4.1) and rearranging gives the equivalent form

$$f(\mathbf{x}) + \nabla f(\mathbf{y})^\top(\mathbf{y} - \mathbf{x}) \geq f(\mathbf{y}) \quad \forall \mathbf{x}, \mathbf{y} \in C.$$

For a strictly convex function and $\mathbf{x} \neq \mathbf{y}$, both displayed inequalities are strict.

Warning

The tangent-plane inequality must hold at every base point $\mathbf{x} \in C$. One global supporting tangent does not imply convexity. For example, define $f : \mathbb{R} \rightarrow \mathbb{R}$ by $f(t) := t^2(t-1)^2$. The tangent at $t = 0$ is the zero function and globally underestimates f , since $f(t) \geq 0$. Nevertheless,

$$f(1/2) = \frac{1}{16} > 0 = \frac{1}{2}f(0) + \frac{1}{2}f(1),$$

so the defining convexity inequality fails.

Example 4.3 (A quadratic verified by its tangent plane). Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $f(x_1, x_2) := x_1^2 + x_2^2$. For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$,

$$f(\mathbf{y}) - f(\mathbf{x}) - \nabla f(\mathbf{x})^\top(\mathbf{y} - \mathbf{x}) = (x_1 - y_1)^2 + (x_2 - y_2)^2 \geq 0.$$

The tangent-plane inequality is strict for $\mathbf{x} \neq \mathbf{y}$, so f is strictly convex.

4.5 First-order optimality over a convex set

Let $U \subseteq \mathbb{R}^n$ be open, let $f : U \rightarrow \mathbb{R}$ be differentiable, and let $\mathbf{x}^* \in U$. For all sufficiently small $\mathbf{h}$, define

$$r_{\mathbf{x}^*}(\mathbf{h}) := f(\mathbf{x}^* + \mathbf{h}) - f(\mathbf{x}^*) - \nabla f(\mathbf{x}^*)^\top \mathbf{h}.$$

Differentiability means

$$\lim_{\mathbf{h} \rightarrow 0} \frac{r_{\mathbf{x}^*}(\mathbf{h})}{\|\mathbf{h}\|} = 0.$$

Thus the sign of $\nabla f(\mathbf{x}^*)^\top \mathbf{h}$ predicts first-order change near $\mathbf{x}^*$. Feasibility determines which small displacements $\mathbf{h}$ may actually be taken.

Theorem 4.5 (Variational inequality). *Let $U \subseteq \mathbb{R}^n$ be open and convex, let $C \subseteq U$ be nonempty and convex, and let $f : U \rightarrow \mathbb{R}$ be convex and continuously differentiable. A point $\mathbf{x}^* \in C$ is a global minimizer of f over C if and only if*

$$\nabla f(\mathbf{x}^*)^\top (\mathbf{x} - \mathbf{x}^*) \geq 0 \quad \forall \mathbf{x} \in C. \quad (4.2)$$

See Figure 4.5 for an illustration.

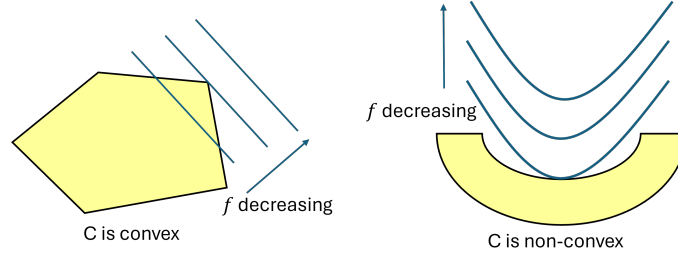


Figure 4.5: At a constrained convex minimizer, the inner product of the gradient with every feasible displacement is nonnegative.

Proof. Suppose first that Equation (4.2) holds. The tangent-plane inequality gives

$$f(\mathbf{x}) \geq f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)^\top (\mathbf{x} - \mathbf{x}^*) \geq f(\mathbf{x}^*) \quad \forall \mathbf{x} \in C,$$

so $\mathbf{x}^*$ is a global minimizer.

Conversely, suppose $\mathbf{x}^*$ is a global minimizer. Fix $\mathbf{x} \in C$ and define

$$\phi(t) := f(\mathbf{x}^* + t(\mathbf{x} - \mathbf{x}^*)), \quad 0 \leq t \leq 1.$$

Convexity of C makes every point on this segment feasible. Since $t = 0$ minimizes ϕ on $[0, 1]$, its right derivative is nonnegative:

$$0 \leq \phi'_+(0) := \lim_{t \downarrow 0} \frac{\phi(t) - \phi(0)}{t} = \nabla f(\mathbf{x}^*)^\top (\mathbf{x} - \mathbf{x}^*).$$

Because $\mathbf{x}$ was arbitrary, Equation (4.2) follows. $\square$

Warning

The two implications use different hypotheses. Convexity of the feasible set supports the necessary condition at a local minimizer even when the objective is not convex. Convexity of the objective makes the variational inequality sufficient for global optimality. For $C := \mathbb{R}$, define $f : \mathbb{R} \rightarrow \mathbb{R}$ by $f(x) := -x^2$. The variational inequality holds at 0, although 0 is a maximizer rather than a minimizer. At a boundary minimizer the gradient also need not vanish; the variational inequality is the appropriate first-order condition.

Corollary 4.6 (Unconstrained convex optimization). *Let $C \subseteq \mathbb{R}^n$ be open and convex, and let $f : C \rightarrow \mathbb{R}$ be convex and continuously differentiable. Then*

$$\mathbf{x}^* \in C \text{ is a global minimizer} \iff \nabla f(\mathbf{x}^*) = \mathbf{0}.$$

If f is strictly convex, this minimizer is unique.

Proof. When C is open, both $\mathbf{x}^* + t\mathbf{d}$ and $\mathbf{x}^* - t\mathbf{d}$ are feasible for all sufficiently small $t > 0$. Applying Equation (4.2) to both directions forces $\nabla f(\mathbf{x}^*)^\top \mathbf{d} = 0$ for every $\mathbf{d}$, hence the gradient is zero. The converse follows from Theorem 4.5. $\square$

Example 4.4 (Boundary and interior minimizers). Define $f_1 : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f_1(x_1, x_2) := (x_1 - 4)^2 + (x_2 - 6)^2.$$

The constrained minimizer from Chapter 2 is $\mathbf{x}^* = (34/13, 66/13)^\top$. Its gradient is

$$\nabla f_1(\mathbf{x}^*) = \begin{bmatrix} -36/13 \\ -24/13 \end{bmatrix} \neq \mathbf{0}.$$

The nonzero gradient is compatible with optimality because $\mathbf{x}^*$ lies on the boundary of the feasible region (see more details in Chapter 12). By contrast, for $f_2 : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $f_2(x_1, x_2) := (x_1 - 2)^2 + (x_2 - 2)^2$, the interior minimizer $(2, 2)^\top$ satisfies $\nabla f_2(2, 2) = \mathbf{0}$.

Chapter summary

- In a convex optimization problem, every local minimizer is global; strict convexity gives uniqueness when a minimizer exists.
- Directional change is $\nabla f(\mathbf{x})^\top \mathbf{d}$. If the gradient is nonzero, its normalized negative is the Euclidean unit direction of steepest descent.
- At a differentiable point with nonzero gradient, the gradient is normal to every level-set tangent traced by a differentiable curve.
- On an open convex domain, a continuously differentiable function is convex exactly when every tangent plane globally underestimates it.
- For a convex differentiable objective over a convex feasible set, global optimality is equivalent to $\nabla f(\mathbf{x}^*)^\top (\mathbf{x} - \mathbf{x}^*) \geq 0$ for every feasible $\mathbf{x}$.

4.6 Exercises

Exercise 4.1 (A nonsymmetric quadratic gradient). Let $A \in \mathbb{R}^{n \times n}$ be arbitrary, and define $q : \mathbb{R}^n \rightarrow \mathbb{R}$ by $q(\mathbf{x}) := \frac{1}{2} \mathbf{x}^\top A \mathbf{x}$. Derive its gradient component by component, and state how the answer differs from the symmetric case.

Exercise 4.2 (Gradient and graphical direction). Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $f(x_1, x_2) := x_1^2/4 + x_2^2$. Compute $\nabla f(2, 1)$, sketch the level curve through $(2, 1)$, and compare the gradient with the direction of increasing contour values.

Exercise 4.3 (Steepest descent direction). Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $f(x_1, x_2) := x_1^2 + 2x_2^2$. Compute the negative-gradient direction at $(1, 1)$, and normalize it to obtain the Euclidean unit direction of steepest descent.

Exercise 4.4 (A perspective function). Define $S := \{(y, z) \in \mathbb{R}^2 : y > 0\}$ and $f : S \rightarrow \mathbb{R}$ by $f(y, z) := y \exp(-z/y)$. Show that f is convex. Conclude that

$$F := \{(x, y, z) \in \mathbb{R}^3 : y \exp(-z/y) \leq x, \ x > 0, \ y > 0\}$$

is convex.

Hint. For another point $(\bar{y}, \bar{z}) \in S$, define $a := -z/y$ and $b := -\bar{z}/\bar{y}$. Use $e^b \geq e^a + e^a(b-a)$ together with [Theorem 4.4](#). For the second part, use [Theorem 3.3](#) after ordering the coordinates as (y, z, x) ; note that $x > 0$ is automatic.